What survives diffusion noise

The EM chapters after the audit: the short version

Brief for the meeting with Jerome and Marc · 20 September 2026

Six pages. Every number is a macro from the frozen runs; the derivations and definitions are in the compendium, the audit trail in the audit report.

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1 The question and the laboratory

A diffusion model learns the score of a noised data law, $\nabla_x \log P_t(x)$, and for real data nobody knows the target. We built a family where it is known exactly and is not trivial: a first-order Markov chain of length L with non-Gaussian innovations (Laplace, Student-t, uniform, bimodal, a one-parameter family in between), noised coordinate by coordinate. Because the noise attaches one factor per site, the posterior stays a chain and two sweeps of exact message passing return the score. The covariance of every innovation family is the same by construction, so the laboratory separates second-order structure (correlation ρ) from higher-order structure (the shape of the innovation law) cleanly. *Compendium, Chapters 3, 6, 10.*

Three questions were asked of it. What information survives the noise, order by order? Can the dynamics be learned from noised sequences alone, and how efficiently? And what does an estimator that knows the structure buy against a neural denoiser that does not, and where does that stop?

2 What stands, in one line each

1. **Noise is a low-pass filter on statistical order.** The Hermite moment of degree m is attenuated exactly by e^{-mt}; the Fisher information about a feature first visible at degree m decays as e^{-2mt}; the number of sequences needed for a fixed local precision grows as e^{2mt}. Correlation lives at degree two, shape at degree four. Verified on the exact score, with the constants. *Compendium, Chapters 2, 4, 9.*

2. **Second-order closure has an exact price.** Gaussian message closure computes the LMMSE of the covariance twin (Proposition 1), and the gap to the true score is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ to leading order. Adding the fourth-order term back closes 0.961 of the gap at $t = 0.68$ and all of it from $t = 2$. *Compendium, Chapters 9, 12.*

3. **The dynamics are learnable through the noise, and the information curve says where learning stops.** From 512 sequences the shape is recovered at the reference up to $t = 0.4$; fits split at $t = 0.7$ (4 of 16 collapse to a Gaussian innovation) and collapse at $t = 1$, one level below the point $t = 0.98$ where the Cramér–Rao reference crosses the coordinate's range. The correlation survives to the crossover. *Compendium, Chapters 4, 14.*

4. **Capacity buys exactly the fourth-order structure.** A Gaussian-innovation kernel lands on the LMMSE floor (0.167 against 0.168); an eight-component mixture reaches 0.023 at $M = 2048$. Across the generalised-Gaussian family the gain is ordered by κ_4 and reverses at the Gaussian member. The contrast holds in all 15 (ρ, L) cells of a regime sweep, and in every cell the $C = 1$ arm lands on the LMMSE floor to 0.0099, which checks the whole apparatus. *Compendium, Chapters 12, 14, 16.*

5. **Against networks, the advantage is largest at equal information.** Under the standard recipe (clean sequences, fresh corruption every step) the best of six architectures reaches 0.060 at $M = 2048$, a paired factor 2.83 behind EM–BP. At EM–BP's own interface (one noised view per sequence, no clean chain, no true score for model selection) the best network reaches 0.165,

a paired factor 7.9 [6.6, 9.3]; given the clean sequence as a target but still one corruption, 0.155, a factor 7.3. *Compendium, Chapters 7, 16, 18.*

6. **The advantage ends where the structure is wrong, at a strength we can compute.** Under an AR(2) coefficient or a site-dependent ρ the ordering reverses inside the swept range; a rank-one global latent or a long-range coupling never reverses it. The large-sample error of the misspecified EM–BP is a closed-form Kullback–Leibler projection bias $b(\varepsilon)$ that reproduces the measured errors to within 0.019. *Compendium, Chapter 17.*

3 What the audit changed

These are corrections to what the 18 September report and the manuscript *said*; no frozen number was wrong.

- **"Information-matched, 7.3×" was the wrong arm.** That network saw the clean chain as a regression target. The matched arm is 7.9×; the 7.3× arm is now reported as "even given the clean sequence".
- **"Attains the Cramér–Rao bound at 1.02×" is withdrawn.** The average mixed levels where the fit is at the reference with levels where it is collapsed; the bound is for a one-parameter family and the estimator is a mixture selected on validation. The paper now reports the per-level picture of item 3 above.
- **"Locality is a ceiling" is withdrawn.** The exact best radius-4 estimator has error 0.030 at $\rho = 0.85$ against the window head's 0.118: the radius costs little, the architecture costs the rest.
- **Theorem 3 needs an exponential moment**, which its proof used and its statement lacked. Restated; every family in the paper satisfies it except Student-t, whose heavy-tailed regime is now a separate proposition. On a wide enough domain the sampled Student-t_5 chain reaches the Hermite constant (1.03 at $t = 4$); its earlier "departure" was grid truncation.
- **"All belief mass outside the domain" was one or two sites in 8192.** The Student-t row of the family ladder is now computed on a $(1601, 16)$ domain, where it has stopped moving (by 0.0002 between the two widest grids).
- **The departure of $\hat{\rho}$ from the reference at $t \geq 1.4$ was partly budget.** With the EM ladder extended to 8000 iterations the error at $t = 1.4$ falls from 0.0325 to 0.0215; at $t = 2$ from 0.3430 to 0.2441 with the ladder still binding in 0.56 of cells. Past the crossover EM is as limited by its convergence as by the information.
- **Model selection without the truth works.** Selecting EM–BP's iteration count on the held-out one-view log-evidence instead of on validation risk against the true kernel gives 0.105 against 0.105 at $M = 32$ (the last rung: 0.208).

4 What is still running, and one thing that went wrong

- **E12b** (sample complexity by order, sixteen seeds, five levels) is two thirds through on the cluster. Its analysis was amended before any data: the target is the exact information curve, whose local slopes on the swept levels are 6–7 for shape and 3.8–3.9 for correlation, not the asymptotic 8 and 4.
- **E14** (the matched comparison across three correlations and two chain lengths) is the claim a referee will press on hardest. Its first runs were void: the driver re-created the random generator inside its sampling loop, so every training chain was the same draw, and both arms fitted 2048 copies of one chain. Found by running two independent training functions on the same data; fixed with an assertion; the corrected run is queued on the GPU partition and takes about a day. Recorded as negative result N-22.

- **The quadrature envelope** now covers the longer chains: the worst drift on the operating grid moves by a factor 1.22 from $L = 32$ to $L = 128$.

5 Decisions for the meeting

1. **Headline.** "Structure beats networks at equal information by 7.9×, and the standard comparison understates it" (recommended), or the standard-recipe factor of 2.83× with the matched one as a robustness check.

2. **Efficiency.** Present learning through noise as "at the reference while identifiable, collapse where the reference says" (recommended) rather than as a Cramér–Rao efficiency statement.

3. **Framing before or after E14.** Wait for the corrected E14 (about a day) before fixing the regime sentence (recommended), or state the headline at one point and add the sweep as it lands.

4. **Page budget.** The ICML body is four pages over. The recorded cut order moves the family figure, the generation table, the length table and the regime paragraph to the appendix; the theory verification is compressed. Agreement needed on what must stay in the body.

5. **Compendium.** A self-contained compendium (twenty chapters: from cumulants and Hermite polynomials to belief propagation, quadrature, EM, quasi-likelihood, score matching and the protocol) is being written so that every line of the paper can be checked without the literature. Its scope and level are for you to set.

Sources: `paper/main.pdf` (manuscript), `reports/audit-2026-09-18.pdf` (the audit: what was found, what was run, what was changed), `reports/compendium/compendium.pdf` (definitions and derivations), `docs/NEGATIVE_RESULTS.md` (N-14 to N-22), `experiments/registry/` (pre-registrations and amendments). Every macro in this brief is regenerated from the frozen outputs by `make check`.